Learning algorithms: subspace methods, PEM, ML

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Outline

- Subspace methods: deterministic case
- Stochastic state-space models as predictors.
- Subspace methods: stochastic case

Recovering Markov parameters: difficulties

Recall: we should eithe assume $M_k = 0$ for $k \geq T$ or compute expectation $E[y(t)u(t-k)]$.

First solution: truncation error

Second solution: we replace $E[y(K)u(K-k)]$ by

$$\frac{1}{\sqrt{N}} \sum_{s=0}^{N} y(s+K)u(s+K-k)$$

but then $N+K \geq K$ Markov parameters enter $y(K+s)$

$$y(K+s) = \sum_{l=0}^{K+s} M_l u(K+s-l)$$

Solution (intuitive): future and past inputs are lin. independent, we can extract lin.comb. of

$$M_{k+l}, \quad l = 0, 1, \ldots,$$

Solution (formal): estimate observability matrix and states

State-space driven by random variables

If the inputs are random variables, so are the outputs

$$\begin{aligned} \mathbf{x}(t+1) &= A\mathbf{x}(t) + B\mathbf{u}(t) \\ \mathbf{y}(t) &= C\mathbf{x}(t) + D\mathbf{u}(t) \end{aligned}$$

observed signals $y(t), u(t), x(t)$ samples of $\mathbf{y}(t), \mathbf{u}(t), \mathbf{x}(t)$.

Data equations

$$\underbrace{\begin{bmatrix} \mathbf{y}(t) \\ \vdots \\ \mathbf{y}(t+k-1) \end{bmatrix}}_{\mathbf{y}_k(t)} = \underbrace{O_k}_{\text{observability matrix}} \mathbf{x}(t) + \underbrace{\Psi_k}_{\text{Toeplitz matrix}} \underbrace{\begin{bmatrix} \mathbf{u}(t) \\ \vdots \\ \mathbf{u}(t+k-1) \end{bmatrix}}_{\mathbf{u}_k(t)}$$

$$O_k = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{k-1} \end{bmatrix}, \quad \Psi_k = \begin{bmatrix} D & 0 & \cdots & 0 & \\ CB & D & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \\ CA^{k-2}B & CA^{k-3}B & \cdots & CB & D \end{bmatrix}$$

Assume:

- the interesection of the linear spaces generated by $\{\mathbf{u}(s)\}_{s<t}$ and $\{\mathbf{u}(s)\}_{s\geq t}$ is trivial ($\{0\}$)
- (A, B, C, D) is minimal
- $k \geq n$, where n is the order of the system.

Basic ideas of subspace methods

1. Recover $O_k, \mathbf{x}(t), \mathbf{x}(t+1)$ from the the random variables $\{\mathbf{y}(t), \mathbf{u}(t)\}_{0 \le t \le t+k+1}$
2. Find A, B, C, D by solving the linear regression

$$\begin{bmatrix} \mathbf{x}(t+1) \\ \mathbf{y}(t) \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{u}(t) \end{bmatrix}$$

Several algorithms: N4SID, MOESP. Differ in details. We concentrate on N4SID.

$O_k \mathbf{x}(t)$ is the *oblique projection* of $\mathbf{y}_k(t)$ onto the past outputs and inputs $\{\mathbf{u}(s), \mathbf{y}(s)\}_{t-k \le s < t}$ along the future inputs $\{\mathbf{u}(s)\}_{s \ge t}$.

Orthogonal projection

$$\mathbf{a} = \begin{bmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_k \end{bmatrix} \in \mathbb{R}^k, \qquad \mathbf{c} = \begin{bmatrix} \mathbf{c}_1 \\ \vdots \\ \mathbf{c}_m \end{bmatrix} \in \mathbb{R}^m$$

Orthogonal projection $E^{\mathbf{c}}\mathbf{a}$ of $\mathbf{a}$ onto the linear space spanned by $\mathbf{c}$:

$$E^{\mathbf{c}}\mathbf{a} \triangleq Z_c\mathbf{c}$$

such that

$$E[(\mathbf{a} - Z_c\mathbf{c})\mathbf{c}^T] = 0$$

It can be shown that

$$Z_c = E[\mathbf{a}\mathbf{c}^T](E[\mathbf{c}\mathbf{c}^T])^{-1}$$

If $\mathbf{a}$ and $\mathbf{c}$ are Gaussian: $E[\mathbf{a} \mid \mathbf{c}] = E^{\mathbf{c}}\mathbf{a}$

Orthogonal projection: examples

Example 1 (Gaussian): Let $\mathbf{a}, \mathbf{c}$ be jointly Gaussian, zero mean.

$$\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

$$\Sigma_{\mathbf{aa}} = \begin{bmatrix} 2 & 0.5 \\ 0.5 & 1 \end{bmatrix}, \quad \Sigma_{\mathbf{cc}} = \begin{bmatrix} 1 & 0.3 \\ 0.3 & 1 \end{bmatrix}, \quad \Sigma_{\mathbf{ac}} = \begin{bmatrix} 0.6 & 0.2 \\ 0.1 & 0.4 \end{bmatrix}$$

Then $Z_c = \Sigma_{\mathbf{ac}}\Sigma_{\mathbf{cc}}^{-1}$ and the projection is

$$E^{\mathbf{c}}\mathbf{a} = Z_c\mathbf{c} = \Sigma_{\mathbf{ac}}\Sigma_{\mathbf{cc}}^{-1}\mathbf{c} = E[\mathbf{a} \mid \mathbf{c}]$$

For Gaussians, orthogonal projection equals conditional expectation.

Example 2 (Uniform): Let $\mathbf{a}, \mathbf{c}$ be independent, uniformly distributed on $[-1, 1]$.
Then $\Sigma_{\mathbf{ac}} = E[\mathbf{a}]E[\mathbf{c}]^T = 0$, so $Z_c = 0$ and $E^{\mathbf{c}}\mathbf{a} = 0$.
The orthogonal projection is zero because $\mathbf{a}$ and $\mathbf{c}$ are independent.

Orthogonal projection: exercises

Exercise 1

Let a and c be jointly Gaussian with $E[a] = E[c] = 0$, $\sigma_a^2 = 2$, $\sigma_c^2 = 1$, $Cov(a, c) = 0.4$.
Compute Z_c and the orthogonal projection $E^c a = Z_c c$. Verify that $E[a - Z_c c] = 0$.

Exercise 2

Let $\mathbf{a} = [a_1, a_2]^T$ and $\mathbf{c} = c_1$ where a_1, a_2, c_1 are independent uniform on $[-1, 1]$.
Show that $Z_c = 0$ and hence $E^{\mathbf{c}}\mathbf{a} = 0$.

Exercise 3

Let c be uniform on $[-1, 1]$ and $a = 0.5c + w$ where w is independent Gaussian noise, $w \sim \mathcal{N}(0, 0.1)$.
Compute the covariances and find Z_c. Compare with the case where $a = 0.5c$ (no noise).

Orthogonal projection: linear regression

$$\mathbf{a} = Z_c \mathbf{c} + Z_e \mathbf{e}$$

and $\mathbf{e}$ and $\mathbf{c}$ are uncorrelated, zero mean, then

$$Z_c = E[\mathbf{a}\mathbf{c}^T](E[\mathbf{c}\mathbf{c}^T])^{-1} \qquad Z_e = E[\mathbf{a}\mathbf{e}^T](E[\mathbf{e}\mathbf{e}^T])^{-1}$$

and

$$E^{\mathbf{c}}\mathbf{a} = Z_c \mathbf{c} = \mathbf{a} - E^{\mathbf{e}}\mathbf{a}$$

Orthogonal projection: linear regression

Example 3 (Linear regression): Let a and $\mathbf{c} = [c_1, c_2]^T$ be jointly Gaussian with $E[a] = E[c_1] = E[c_2] = 0$, $\sigma_a^2 = 1$, $\sigma_{c_1}^2 = \sigma_{c_2}^2 = 1$, and $\mathrm{Cov}(a, c_1) = 0.6$, $\mathrm{Cov}(a, c_2) = 0.3$, $\mathrm{Cov}(c_1, c_2) = 0.2$.
Compute $Z_c = [z_1, z_2]$ such that $E^{\mathbf{c}}a = z_1 c_1 + z_2 c_2$ and find $\Sigma_{\mathbf{cc}} = E[\mathbf{cc}^T]$, then $Z_c = E[a\mathbf{c}^T]E[\mathbf{cc}^T]^{-1}$.

Linear regression: exercises

Exercise 4

Let a, c_1, c_2 *be independent uniform on* $[-1, 1]$ *with* $E[a] = E[c_1] = E[c_2] = 0$.
Show that $\Sigma_{a,\mathbf{c}} = 0$ *and hence the orthogonal projection* $E^{\mathbf{c}}a = 0$.
Interpret this result: what does it mean if a *and* $\mathbf{c}$ *are independent?*

Exercise 5

Let c_1, c_2 *be independent uniform on* $[-1, 1]$ *and* $a = 0.3c_1 + 0.5c_2 + w$ *where* $w \sim \mathcal{N}(0, 0.2)$ *is independent of* c_1, c_2.
Compute $\Sigma_{\mathbf{cc}}$, $E[a\mathbf{c}^T]$, *and find* $Z_c = [z_1, z_2]$. *Compare the values* z_1, z_2 *with the true coefficients* $0.3, 0.5$.

Exercise 6

$\Sigma_{\mathbf{aa}} = E[\mathbf{aa}^T] = I_2$, $\Sigma_{\mathbf{cc}} = E[\mathbf{cc}^T] = I_2$, *and* $\Sigma_{\mathbf{ac}} = E[\mathbf{ac}^T] = \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.6 \end{bmatrix}$.
Compute the orthogonal projection matrix Z_c *and find* $E^{\mathbf{c}}\mathbf{a}$.

Subspace method: white noise input

If $\mathbf{u}$ white noise, then $\mathbf{x}(t)$ is a linear sum

$$\mathbf{x}(t) = \sum_{s=0}^{t-1} A^{t-s-1} B\mathbf{u}(s) = A^k \mathbf{x}(t-k) + \sum_{s=t-k}^{t-1} A^{t-s-1} B\mathbf{u}(s)$$

and hence uncorrelated with $\mathbf{u}_k(t)$.

$$\begin{aligned}
\mathbf{y}_k(t-k) = O_k \mathbf{x}(t-k) + \Psi_k \mathbf{u}_k(t-k) &\implies \\
\mathbf{x}(t-k) = O_k^{+}(\mathbf{y}_k(t-k) - \Psi_k \mathbf{u}_k(t-k)) &\implies \\
\mathbf{x}(t) = A^k \mathbf{x}(t-k) + \sum_{s=t-k}^{t-1} A^{t-s-1} B\mathbf{u}(s) &= \\
\underbrace{Z_{\mathbf{y},\mathbf{u}}}_{\text{some matrix}} \underbrace{\begin{bmatrix} \mathbf{y}_k(t-k) \\ \mathbf{u}_k(t-k) \end{bmatrix}}_{\mathbf{p}_k(t)} &
\end{aligned}$$

Subspace method: white noise input

$$\mathbf{y}_k(t) = O_k \underbrace{Z_{\mathbf{y},\mathbf{u}} \mathbf{p}_k(t)}_{\mathbf{x}(t)} + \Psi_k \mathbf{u}_k(t)$$

Note that $\mathbf{p}_k(t)$ is uncorrelated with $\mathbf{u}_k(t)$.

$$O_k \mathbf{x}(t) = E^{\mathbf{p}_k(t)} \mathbf{y}_k(t)$$
$$O_k \mathbf{x}(t+1) = E^{\mathbf{p}_k(t+1)} \mathbf{y}_k(t+1)$$

If we perform SVD on

$$E[\mathbf{y}_k(t)\mathbf{p}_k(t)^T] E[\mathbf{p}_k(t)\mathbf{p}_k(t)^T]^{-1} = USV^T \implies$$
$$O_k = US^{1/2}, \quad \mathbf{x}(t) = S^{1/2} V^T \mathbf{p}_k(t)$$
$$\mathbf{x}(t+1) = O_k^{\dagger} E^{\mathbf{p}_k(t+1)} \mathbf{y}_k(t+1)$$

up to change of basis.

What if $\mathbf{u}$ is not white noise ?

Oblique projection

Random variables

$$\mathbf{a} = \begin{bmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_k \end{bmatrix} \in \mathbb{R}^k, \qquad \mathbf{b} = \begin{bmatrix} \mathbf{b}_1 \\ \vdots \\ \mathbf{b}_l \end{bmatrix} \in \mathbb{R}^l, \quad \mathbf{c} = \begin{bmatrix} \mathbf{c}_1 \\ \vdots \\ \mathbf{c}_m \end{bmatrix} \in \mathbb{R}^m$$

Assume: only trivial intersection between the rows spaces of **b** and **c**:

$$A\mathbf{b} + B\mathbf{c} = 0 \implies A\mathbf{b} = 0, B\mathbf{c} = 0$$

and

$$\mathbf{a} = Z_b\mathbf{b} + Z_c\mathbf{c}$$

Oblique projection $E^{\mathbf{b}}_{\|\mathbf{c}}\mathbf{a}$ of **a** onto the linear space spanned by **b** along the linear space spanned by **c**:

$$E^{\mathbf{b}}_{\|\mathbf{c}}\mathbf{a} \triangleq Z_b\mathbf{b}$$

Oblique projection: special case

If **c** and **b** are uncorrelated

$$E^{\mathbf{b}}_{\|\mathbf{c}}\mathbf{a} = E^{\mathbf{b}}\mathbf{a}$$

Can be shown that

$$Z_b = \Sigma_{\mathbf{ab}|\mathbf{c}}\Sigma_{\mathbf{bb}|\mathbf{c}}^{-1}$$

$\Sigma_{\mathbf{ab}|\mathbf{c}}$ and $\Sigma_{\mathbf{bb}|\mathbf{c}}$ are the conditional covariances of **a** and **b** given **c**

$$\Sigma_{\mathbf{ab}|\mathbf{c}} = E[(\mathbf{a} - E^{\mathbf{c}}\mathbf{a})(\mathbf{b} - E^{\mathbf{c}}\mathbf{b})^T] = \Sigma_{\mathbf{ab}} - \Sigma_{\mathbf{ac}}\Sigma_{\mathbf{cc}}^{-1}\Sigma_{\mathbf{cb}}$$
$$\Sigma_{\mathbf{bb}|\mathbf{c}} = E[(\mathbf{b} - E^{\mathbf{c}}\mathbf{b})(\mathbf{b} - E^{\mathbf{c}}\mathbf{b})^T] = \Sigma_{\mathbf{bb}} - \Sigma_{\mathbf{bc}}\Sigma_{\mathbf{cc}}^{-1}\Sigma_{\mathbf{cb}}$$

Oblique projection: examples

Example 1: Let $\mathbf{a} = [a_1, a_2]^T$, $\mathbf{b} = [b_1, b_2]^T$, $\mathbf{c} = [c_1, c_2]^T$ be jointly Gaussian with zero mean.

$$\Sigma_{\mathbf{aa}} = \begin{bmatrix} 1 & 0.3 \\ 0.3 & 1 \end{bmatrix}, \quad \Sigma_{\mathbf{bb}} = \begin{bmatrix} 2 & 0.4 \\ 0.4 & 2 \end{bmatrix}, \quad \Sigma_{\mathbf{cc}} = \begin{bmatrix} 1.5 & 0.2 \\ 0.2 & 1.5 \end{bmatrix}$$

$$\Sigma_{\mathbf{ab}} = \begin{bmatrix} 0.5 & 0.2 \\ 0.3 & 0.4 \end{bmatrix}, \quad \Sigma_{\mathbf{ac}} = \begin{bmatrix} 0.3 & 0.1 \\ 0.2 & 0.3 \end{bmatrix}$$

Compute the conditional covariance $\Sigma_{\mathbf{ab|c}} = \Sigma_{\mathbf{ab}} - \Sigma_{\mathbf{ac}}\Sigma_{\mathbf{cc}}^{-1}\Sigma_{\mathbf{cb}}$ and the oblique projection $E^{\mathbf{b}}_{\|\mathbf{c}}\mathbf{a}$.

Example 2: If $\mathbf{a}$ and $\mathbf{b}$ are independent (uncorrelated), show that $E^{\mathbf{b}}_{\|\mathbf{c}}\mathbf{a} = -\Sigma_{\mathbf{ac}}\Sigma_{\mathbf{cc}}^{-1}\Sigma_{\mathbf{cb}}\Sigma_{\mathbf{bb|c}}^{-1}\mathbf{b}$.

Oblique projection: exercises

Exercise 7

Let $\mathbf{a} = [a_1, a_2]^T$, $\mathbf{b} = [b_1]$, $\mathbf{c} = [c_1, c_2]^T$ *be jointly Gaussian with zero mean,* $\Sigma_{\mathbf{aa}} = I_2$, $\sigma_b^2 = 1$, $\Sigma_{\mathbf{cc}} = I_2$, $\Sigma_{\mathbf{ab}} = [0.6, 0.3]^T$, $\Sigma_{\mathbf{ac}} = \begin{bmatrix} 0.4 & 0.2 \\ 0.3 & 0.5 \end{bmatrix}$, $\sigma_{\mathbf{bc}} = [0.2, 0.4]^T$.
Compute $Z_b = \Sigma_{\mathbf{ab}|\mathbf{c}} \Sigma_{\mathbf{bb}|\mathbf{c}}^{-1}$ *and the oblique projection* $E_{\|\mathbf{c}}^{\mathbf{b}} \mathbf{a} = Z_b \mathbf{b}$.

Exercise 8

Verify that the oblique projection satisfies $\mathbf{a} - E_{\|\mathbf{c}}^{\mathbf{b}} \mathbf{a}$ *is uncorrelated with* $\mathbf{b}$ *when conditioned on* $\mathbf{c}$.

Exercise 9

Let $\mathbf{a}, \mathbf{b}$ *be independent uniform on* $[-1, 1]$ *and* $\mathbf{c} = \mathbf{a} + \mathbf{b} + w$ *where* $w \sim \mathcal{N}(0, 0.5)$.
Show that $E_{\|\mathbf{c}}^{\mathbf{b}} \mathbf{a} \neq 0$ *even though* $\mathbf{a}$ *and* $\mathbf{b}$ *are independent, because they are both correlated with* $\mathbf{c}$.

Subspace method: general case

As before

$$\mathbf{x}(t) = Z_{\mathbf{y},\mathbf{u}} \begin{bmatrix} \mathbf{y}_k(t-k) \\ \mathbf{u}_k(t-k) \end{bmatrix} = Z_{\mathbf{y},\mathbf{u}} \mathbf{p}_k(t)$$

$\mathbf{p}_k(t)$ - inputs and outputs before t (past)

$$\mathbf{y}_k(t) = O_k \mathbf{x}(t) + \Psi_k \mathbf{u}_k(t) = O_k Z_{\mathbf{y},\mathbf{u}} \mathbf{p}_k(t) + \Psi_k \mathbf{u}_k(t)$$

$\mathbf{y}_k(t), \mathbf{u}_k(t)$ - outputs and inputs after t (future).

$$O_k \mathbf{x}(t) = E^{\mathbf{p}_k(t)}_{\|\mathbf{u}_k(t)} \mathbf{y}_k(t) = \Sigma_{\mathbf{y}_k(t)\mathbf{p}_k(t)|\mathbf{u}_k(t)} \Sigma^{-1}_{\mathbf{p}_k(t)\mathbf{p}_k(t)|\mathbf{u}_k(t)} \mathbf{p}_k(t)$$

– oblique projection of future outputs to past outputs and inputs along future inputs.

Subspace method: summary of steps

1. Compute conditional covariance

$$\Sigma_1 = \Sigma_{\mathbf{y}_k(t)\mathbf{p}_k(t)|\mathbf{u}_k(t)}$$
$$\Sigma_2 = \Sigma_{\mathbf{y}_k(t+1)\mathbf{p}_k(t+1)|\mathbf{u}_k(t+1)}$$

2. Factorize (economic SVD), $S = \mathrm{diag}(s_1, \ldots, s_n)$, $n = \mathrm{rank}\ S$, L, M invertable matrices,
$$L\Sigma_1 M = USV^T$$
3. $x(t) = S^{1/2} V^T M^{-1} \Sigma^{-1}_{\mathbf{p}_k(t)\mathbf{p}_k(t)|\mathbf{u}_k(t)} \mathbf{p}_k(t)$, $O_k = US^{1/2}$
4. $\bar{O}_k$ – the first pn rows of O_k.
5. $x(t+1) = \bar{O}_k^{\dagger} L \Sigma_2 \Sigma^{-1}_{\mathbf{p}_k(t+1)\mathbf{p}_k(t+1)|\mathbf{u}_k(t+1)} \mathbf{p}_k(t+1)$.
6. Solve linear least squares

$$\begin{bmatrix} x(t+1) \\ y(t) \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} x(t) \\ u(t) \end{bmatrix}$$

Implementation in terms of data

Identify random variable $\mathbf{a}(t)$ with the column vector of observations from a sample $a(t)$.

$$a_N(t) = \begin{bmatrix} a(t) & a(t+1) & \cdots & a(t+N-1) \end{bmatrix}$$

Assume $\mathbf{a}(t)$ is zero mean, wide-sense stationary, and the law of large numbers holds for mean and variance of $\mathbf{a}_N(t)$:

$$\lim_{N\to\infty} \frac{1}{N} a_N(t) a_N(t)^T = E[\mathbf{a}(t)\mathbf{a}(t)^T]$$

$$\mathbf{z}(t) = E^{\mathbf{b}(t)}\mathbf{a}(t) \approx z_N(t) = a_N(t) b_N^T(t)(b_N(t) b_N^T(t))^{-1} b_N(t)$$

i.e., orthogonal project is represented by the least squares solution of

$$z_N(t) = \min_Z \|a_N(t) - Z b_N(t)\|_F^2$$

Subspace method: from random variables to data

Oblique projection:
row spaces of (t) and $\mathbf{c}(t)$ have trivial intersection $\iff$

$$\text{row span}(b_N(t)) \cap \text{row span}(c_N(t)) = \{0\}$$

Computing oblique projection:

$$\Sigma_{\mathbf{a}(t)\mathbf{b}(t)|\mathbf{c}(t)} \approx \frac{1}{\sqrt{N}}(a_N(t)b_N^T(t) - a_N(t)c_N^T(t)(c_N(t)c_N^T(t))^{-1}c_N(t)b_N^T(t))$$

$$E_{\|\mathbf{c}(t)}^{\mathbf{b}(t)}\mathbf{a}(t) \approx \frac{1}{\sqrt{N}}(a_N(t)b_N^T(t) - a_N(t)c_N^T(t)(c_N(t)c_N^T(t))^{-1}c_N(t)b_N^T(t))b_N(t)$$

Orthogonal and oblique projections for matrices

A, B, C matrices with N columns.

Orthogonal projection of rows of A onto rowspace of B

$$P^B A = AB^T(BB^T)^{-1}B, \quad P^B A = \min_Z \|A - ZB\|_F^2$$

Oblique projection of the rows of A onto B along the row space C

$$P^B_{\|C} A = \Sigma_{A,B|C}\Sigma^{-1}_{B,B|C} B$$
$$\Sigma_{A,B|C} = (A - P^C A)(B - P^C B)^T$$

orthoginal/oblique projections of random variables $\iff$
orthogonal/oblique projection of column vectors formed by their samples.

Data matrices for subspace identification

$$\mathbf{y}_k(t) \approx Y_{t|k} = \begin{bmatrix} y(t) & y(t+1) & \cdots & y(t+N) \\ y(t+1) & y(t+2) & \cdots & y(t+N+1) \\ \vdots & \vdots & \ddots & \vdots \\ y(t+k-1) & y(t+k-1) & \cdots & y(t+N+k-2) \end{bmatrix}$$

$$\mathbf{u}_k(t) \approx U_{t|k} = \begin{bmatrix} u(t) & u(t+1) & \cdots & u(t+N) \\ u(t+1) & u(t+2) & \cdots & u(t+N+1) \\ \vdots & \vdots & \ddots & \vdots \\ u(t+k-1) & u(t+k-1) & \cdots & u(t+N+k-2) \end{bmatrix}$$

$$\mathbf{p}_k(t) \approx W_{t|k} = \begin{bmatrix} Y_{t-k|k} \\ U_{t-k|k} \end{bmatrix}$$

$$\mathbf{x}_k(t) \approx X_t = \begin{bmatrix} x(t) & x(t+1) & \cdots & x(t+N) \end{bmatrix}$$

$$Y_{t|k} = O_k X_t + \Psi_k U_{t|k} = O_k Z_{\mathbf{y},\mathbf{u}} W_{t|k} + \Psi_k U_{t|k}$$

$O_k X_t$ – oblique projection of $Y_{t|k}$ onto the row space of $W_{t|k}$ along the row space of $U_{t|k}$.

Subspace identification with data matrices

1. Compute conditional covariance

$$\Sigma_1 = \Sigma_{Y_{t|k} W_{t|k} | U_{t|k}}$$

$$\Sigma_2 = \Sigma_{Y_{t+1|k} W_{t+1|k} | U_{t+1|k}}$$

2. Factorize (economic SVD), $S = \mathrm{diag}(s_1, \ldots, s_n)$, $n = \mathrm{rank}\ S$, L, M invertable matrices (e.g., $M^{-T} M^{-1} = \Sigma_{W_{t|k} W_{t|k} | U_{t|k}}$, $L^{-1} L^{-T} = \Sigma_{Y_{t|k} Y_{t|k} | U_{t|k}}$)

$$L \Sigma_1 M = U S V^T$$

3. $X_t = S^{1/2} V^T M^{-1} \Sigma^{-1}_{W_{t|k} W_{t|k} | U_{t|k}} W_{t|k}$, $O_k = U S^{1/2}$
4. $\bar{O}_k$ – the first pn rows of O_k.
5. $X_{t+1} = \bar{O}_k^{\dagger} L \Sigma_2 \Sigma^{-1}_{W_{t+1|k} W_{t+1|k} | U_{t+1|k}} W_{t+1|k}$.
6. Solve linear least squares

$$\begin{bmatrix} X_{t+1} \\ Y_{t|k} \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} X_t \\ U_{t|k} \end{bmatrix}$$

i.e.

$$A, B, C, D = \min_Z \left\| \begin{bmatrix} X_{t+1} \\ Y_{t|k} \end{bmatrix} - Z \begin{bmatrix} X_t \\ U_{t|k} \end{bmatrix} \right\|_F^2$$

Subspace identification: correctness

If $N, k \to \infty$, then subspace methods returns $(\hat{A}, \hat{B}, \hat{C}, \hat{D})$ s.t.

$$(T\hat{A}T^{-1}, T\hat{B}, \hat{C}T^{-1}, \hat{D}) = (A, B, C, D)$$

Exercise 10

Consider a 2×2 stable system in controllable canonical form:

$$A = \begin{bmatrix} 0 & 1 \\ -a_1 & -a_2 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} c_1 & c_2 \end{bmatrix}, \quad D = 0$$

with eigenvalues $\lambda_1 = 0.5, \lambda_2 = 0.3$ (so $a_1 = 0.15, a_2 = 0.8$).
Choose $c_1 = 1, c_2 = 0.5$, and let $F = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$, $H = 0.1$.

Is the system stable.

1. *Generate $N = 1000$ samples of data $\{y(t), u(t)\}$ from the system with white noise input $u(t) \sim \mathcal{N}(0,1)$ and $v(t) \sim \mathcal{N}(0,1)$.*
2. *Apply the N4SID algorithm to estimate $\hat{A}, \hat{B}, \hat{C}, \hat{D}$.*
3. *Compute the one-step ahead predictions $\hat{y}(t)$ using both the original system and the estimated system, and compare the prediction error.*
4. *Find the state transformation matrix T such that $T\hat{A}T^{-1} \approx A$, $T\hat{B} \approx B$, $\hat{C}T^{-1} \approx C$.*